

MAT 3150 — Homework 3

Problem 3 Walkthrough: $\infty + \text{Bounded} = \infty$

Goal. Prove that if $\lim_{n \rightarrow \infty} x_n = \infty$ and $(y_n)_n$ is bounded, then $\lim_{n \rightarrow \infty} (x_n + y_n) = \infty$.

The definitions

Bounded. (y_n) is bounded if there is a number $M > 0$ with

$$|y_n| \leq M \quad \text{for every } n.$$

Unpacking the absolute value: $-M \leq y_n \leq M$. For this problem you only need the *lower* half, $y_n \geq -M$. (Why the lower half? We want $x_n + y_n$ to be *big*, so the danger is y_n being very negative.)

Diverging to ∞ . $a_n \rightarrow \infty$ means:

For every $K \in \mathbb{R}$ there is $N \in \mathbb{N}$ such that $a_n > K$ for all $n \geq N$.

Same game as the ε definition, but now the challenger hands you a *height* K (maybe huge, like 10^6), and you must find a point N after which every term is above K .

The key move for this whole topic. The hypothesis $x_n \rightarrow \infty$ works for *every* height. So you are free to apply it to a *different* height than K — whatever height makes your target work out.

Worked example: if $x_n \rightarrow \infty$, then $3x_n - 5 \rightarrow \infty$

Scratch work. Given a height K , we want $3x_n - 5 > K$. Solve for x_n :

$$3x_n - 5 > K \iff x_n > \frac{K+5}{3}.$$

So we should apply the hypothesis to the height $\frac{K+5}{3}$, not to K .

The proof.

Proof. Let $K \in \mathbb{R}$. Since $x_n \rightarrow \infty$, applying the definition with the height $\frac{K+5}{3}$ gives $N \in \mathbb{N}$ such that

$$x_n > \frac{K+5}{3} \quad \text{for all } n \geq N.$$

Then for all $n \geq N$,

$$3x_n - 5 > 3 \cdot \frac{K+5}{3} - 5 = K.$$

Since K was arbitrary, $3x_n - 5 \rightarrow \infty$. □

Notice: the proof starts with “Let K ” (the height for the *conclusion*), then feeds a *modified* height into the *hypothesis*.

Your turn: $x_n \rightarrow \infty$ and (y_n) bounded $\Rightarrow x_n + y_n \rightarrow \infty$

1. Write down what you're given.

- Bounded: there is $M > 0$ with _____ for every n .
In particular, $y_n \geq$ _____ for every n .
- $x_n \rightarrow \infty$: for every height H there is N with _____ for all $n \geq N$.

2. Write down what you must show. For every K there is N such that _____ $> K$ for all $n \geq N$.

3. Scratch work: get rid of y_n . Using $y_n \geq -M$, find a lower bound for $x_n + y_n$ that involves only x_n and M :

$$x_n + y_n \geq \text{_____}.$$

For this lower bound to be bigger than K , you need

$$x_n > \text{_____}.$$

That is the height to feed into the hypothesis.

4. The proof. Fill in the template:

Since (y_n) is bounded, there is $M > 0$ with $|y_n| \leq M$, so in particular $y_n \geq -M$ for all n .

Let $K \in \mathbb{R}$. Since $x_n \rightarrow \infty$, applying the definition with the height _____ gives $N \in \mathbb{N}$ such that

$$x_n > \text{_____} \quad \text{for all } n \geq N.$$

Then for all $n \geq N$,

$$x_n + y_n \geq \text{_____} > \text{_____} = K.$$

Since K was arbitrary, $x_n + y_n \rightarrow \infty$.

Check yourself.

- Why is the first step “ $\geq$ ” and not “ $>$ ”?
- Which step uses boundedness, and which uses $x_n \rightarrow \infty$?
- Where did you use that K was arbitrary?